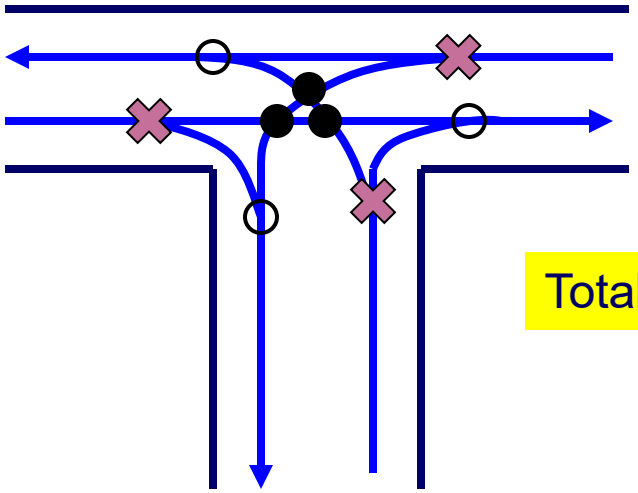
Learning Objectives

- Identify conflict points at an intersection
- Understand MUTCD conditions that warrant a yield, 2-way stop, multi-way stop, and signal control.
- Know how to evaluate adequacy of sight distance of uncontrolled and 2-way stop controlled intersections.

Conflict points at T-intersection



Total = 9

● Crossing = 3

○ Merging = 3

✖ Diverging = 3

Hierarchy of intersection control

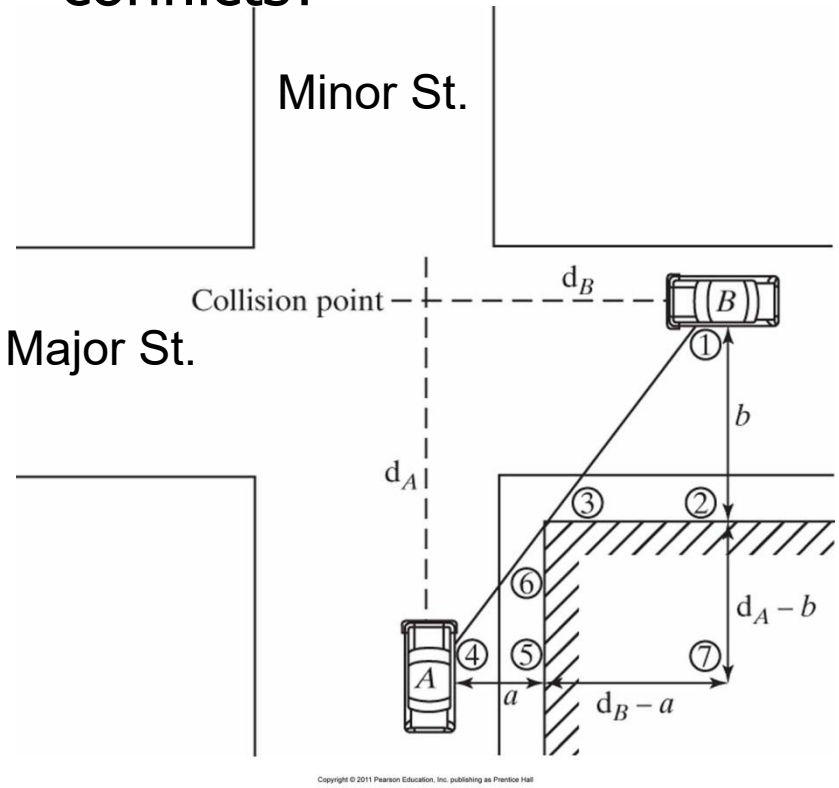
- Level I: basic rules of the road
- Level II: direct assignment of right-of-way using Yield or Stop signs
- Level III: traffic signalization
- The selection of an appropriate level depends on the ability of the drivers to see potential conflicts at the intersection (i.e. sight distance) and traffic volume.
- When it is not reasonable to expect a driver to perceive and avoid a particular conflict, traffic controls must be provided.

Level I control: basic rules of the road

- Basic rules of the road apply at any intersection where right-of-way is not explicitly assigned through the use of traffic signals, stop, or yield signs.
- "**A motorist arriving at the intersection must yield the right of way to motorists who arrived before him, waiting his turn to enter the intersection. If two motorists arrive at the same time, if on different streets, the driver on the left should yield to the driver on the right; or if on the same street, a driver desiring to turn left should yield to the driver from the opposite direction**." *SC Dept. of Motor Vehicle*
- To safely operate under basic rules of the road, drivers on conflicting approaches must be able to see each other in time to assess whether they will collide and take appropriate action. Sight distance must be analyzed to ensure that they are sufficient for drivers to judge and avoid conflicts.

Viability of basic rules-of-the-road

- What distances are needed for drivers to avoid conflicts?



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AASHTO Guidelines: both vehicles must have at least one safe SSD to the collision point.

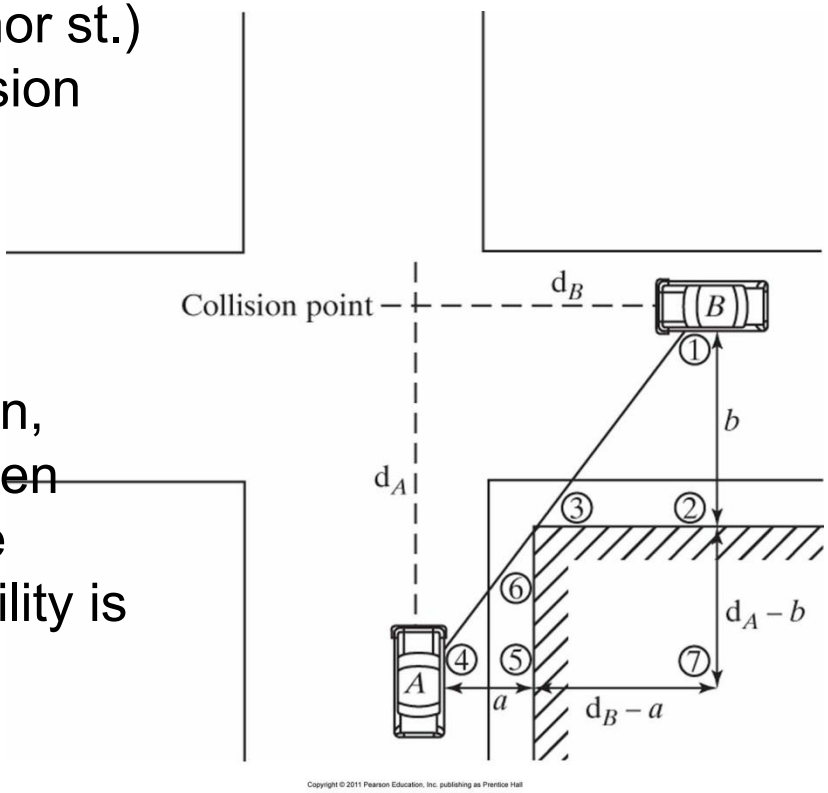
Procedure for determining if an intersection's sight distance is adequate

Step 1: Assume that Vehicle A (on minor st.) is located one safe SSD from the collision point.

$$d_A = 1.47 S_A t + \frac{S_A^2}{30(f \pm G)}$$

Step 2: Based on the assumed position, determine the location of Vehicle B when drivers first see each other. This is the actual position of Vehicle B when visibility is established.

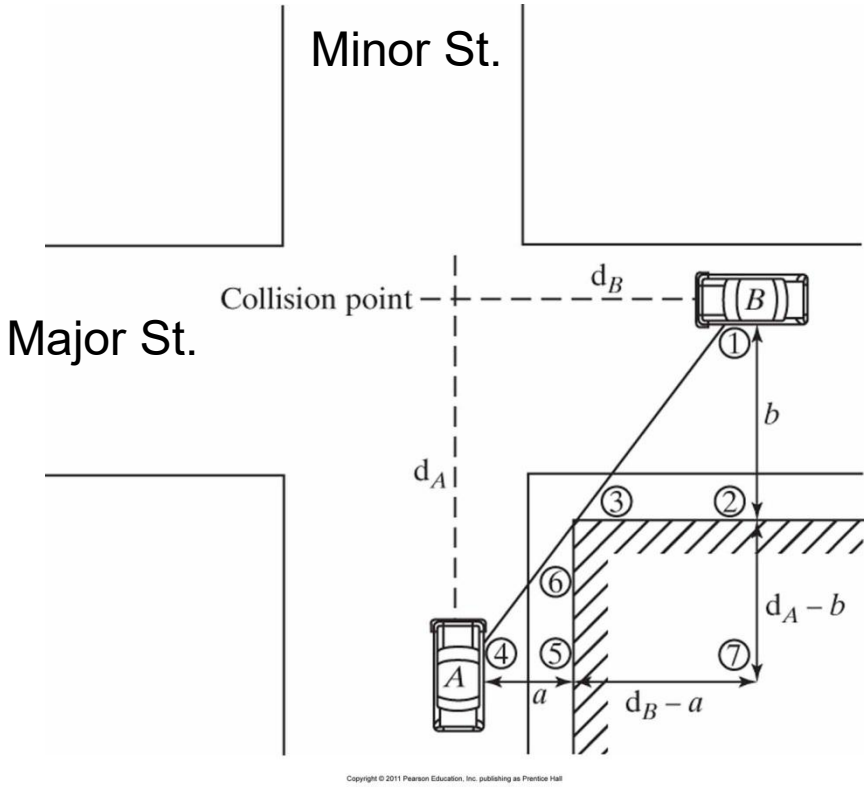
$$d_{B(act)} = \frac{a d_A}{d_A - b}$$



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From the similarity of triangles, Δ321 to Δ471:

$$\frac{b}{d_B - a} = \frac{d_A - b}{a}$$



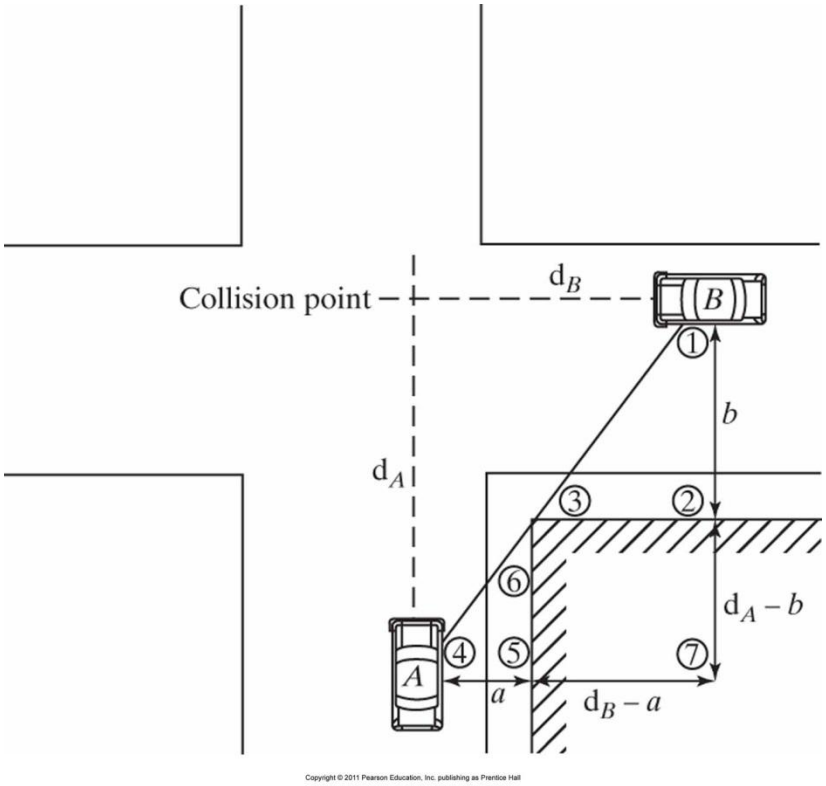
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Procedure for determining if an intersection's sight distance is adequate

Step 3: AASHTO requires Vehicle B to have one safe SSD:

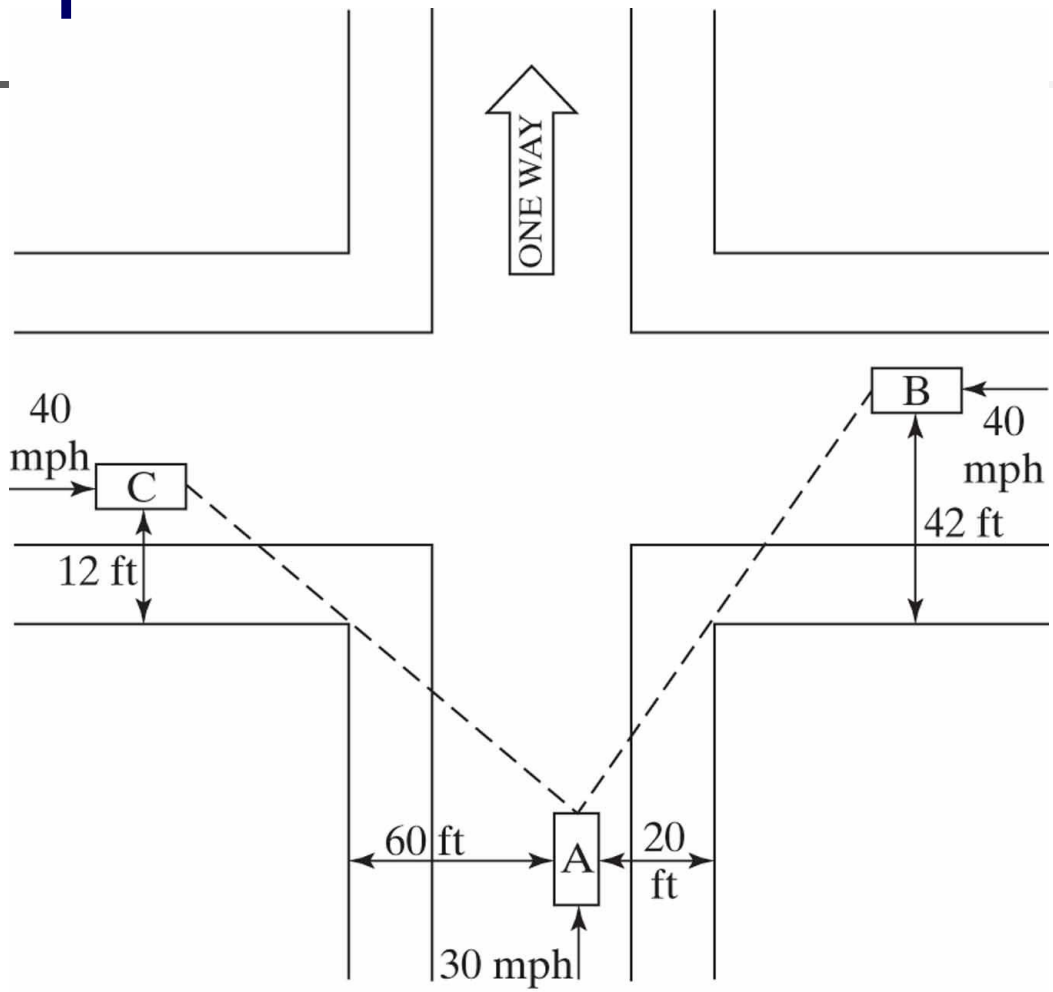
$$d_{B(\min)} = 1.47 S_B t + \frac{S_B^2}{30(f \pm G)}$$

Step 4: If $d_{B(act)} \geq d_{B(min)}$, then adequate SSD for basic rules-of-the-road has been provided.



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Example



(reaction time = 2.5 s, level grade)

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Example

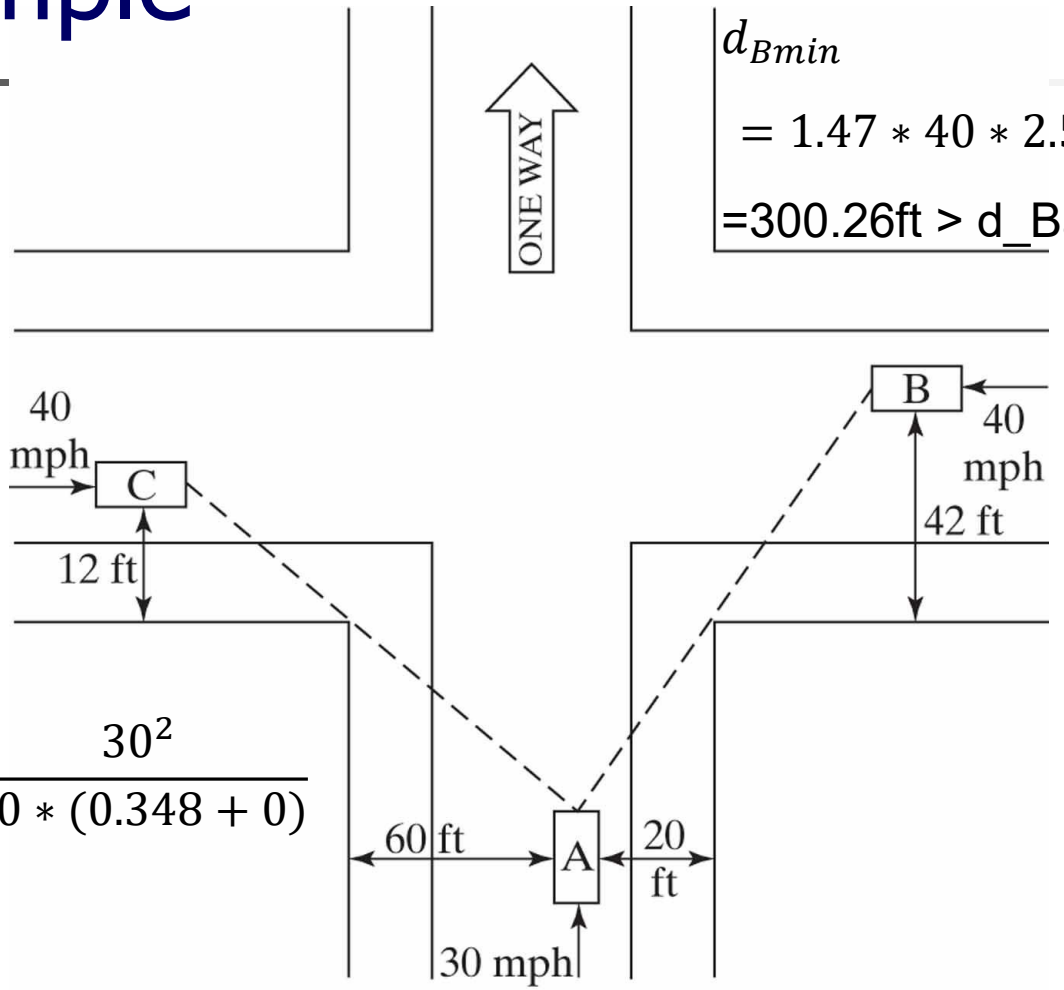
Minor St.

$$d_{Bact} = \frac{ad_A}{d_A - b} = \frac{20 * 196.46}{196.46 - 42} = 25.44ft$$

$$d_{Bmin} = 1.47 * 40 * 2.5 + \frac{40^2}{30 * (0.348 + 0)}$$

=300.26ft > d_Bact

Major St.



(reaction time = 2.5 s, level grade)

$$d_A = 1.47 * 30 * 2.5 + \frac{30^2}{30 * (0.348 + 0)}$$

=196.46 ft

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Potential countermeasures when $d_{b(act)} < d_{b(min)}$?

- Implement intersection control, using Stop or Yield control, or traffic signals.
- Lower the speed limit on the major street to a point where sight distance are adequate.
- Remove or reduce sight obstruction to provide adequate sight distances.

Level II control: yield and stop control

- If the intersection sight triangle is inadequate, then some form of level II control is needed.
- Even if sight triangle is adequate, it may be necessary to implement a higher level of control due to traffic volume and/or complexity of the intersection environment.

Warrants for using 2-way stop or yield control at an intersection (Section 2B.04 Right-of-Way at Intersections)

- Guidance:
- YIELD or STOP signs should be used at an intersection if one or more of the following conditions exist:

A. An intersection of a less important road with a main road where application of the normal right-of-way rule would not be expected to provide reasonable compliance with the law;

B. A street entering a designated through highway or street; and/or

C. An unsignalized intersection in a signalized area.

Warrants for stop signs (Section 2B.06 STOP Sign Applications)

- Guidance:
- The use of STOP signs on the minor-street approaches *should* be considered if engineering judgment indicates that a **stop is always required** because of one or more of the following conditions:

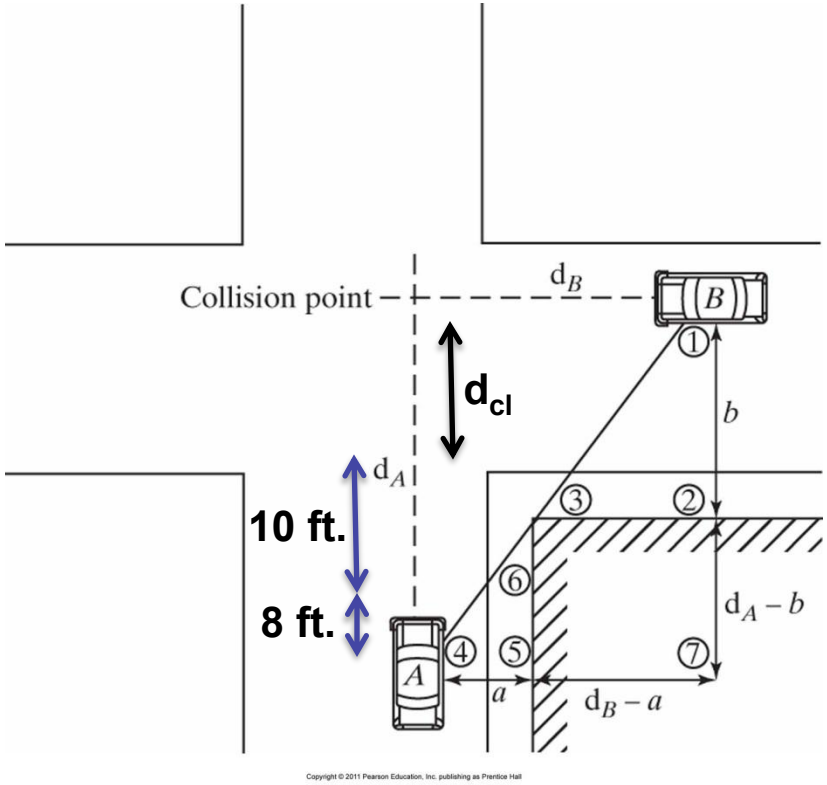
A. The **vehicular traffic volumes** on the through street or highway **exceed 6,000 vehicles per day**;

B. A **restricted view exists** that requires road users to stop in order to adequately observe conflicting traffic on the through street or highway; and/or

C. **Crash records** indicate that three or more crashes that are susceptible to correction by the installation of a STOP sign have been reported within a 12-month period, or that five or more such crashes have been reported within a 2-year period. Such crashes include right-angle collisions involving road users on the minor-street approach failing to yield the right-of-way to traffic on the through street or highway.

Sight distance criteria for stop-controlled intersections

- AASHTO recommends a methodology based on observed gap acceptance behavior of drivers at stop-controlled intersections, assuming
  - Distance from driver's eye to front of vehicle, 8ft
  - Distance from the front of the vehicle to the curb line, 10ft



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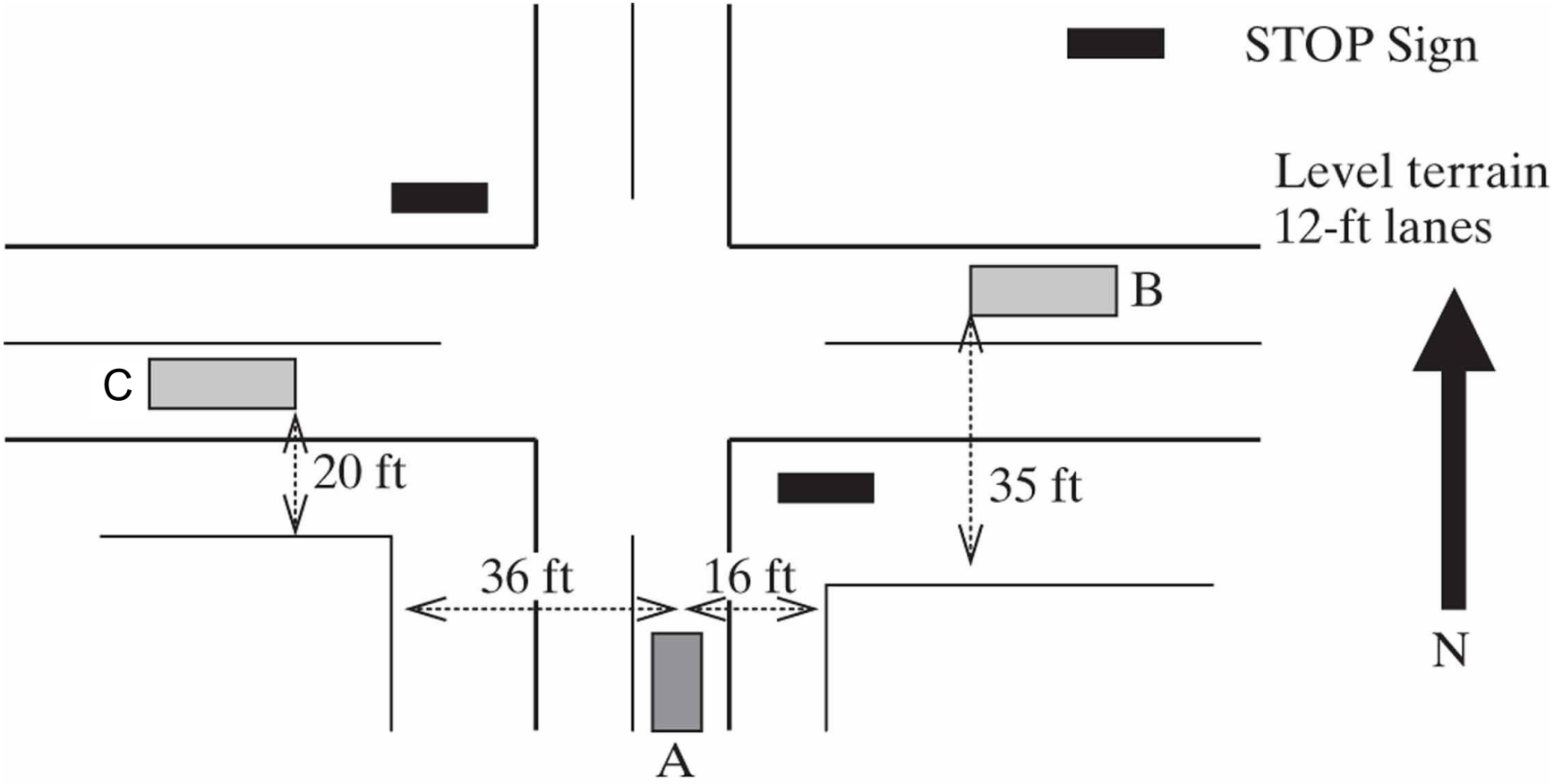
d_cl = distance from the curb line to the center-line of the closest travel lane from the direction under consideration, ft
S_maj = designed speed of major street, mph
T_g = average gap accepted by the minor street driver to enter the major road, s, ***7.5s***

$$d_{A-STOP} = 18 + d_{cl}$$

$$d_{B(\min)} = 1.47 \times S_{maj} \times t_g$$

$$d_{Bact} = \frac{a d_{A-STOP}}{d_{A-STOP} - b}$$

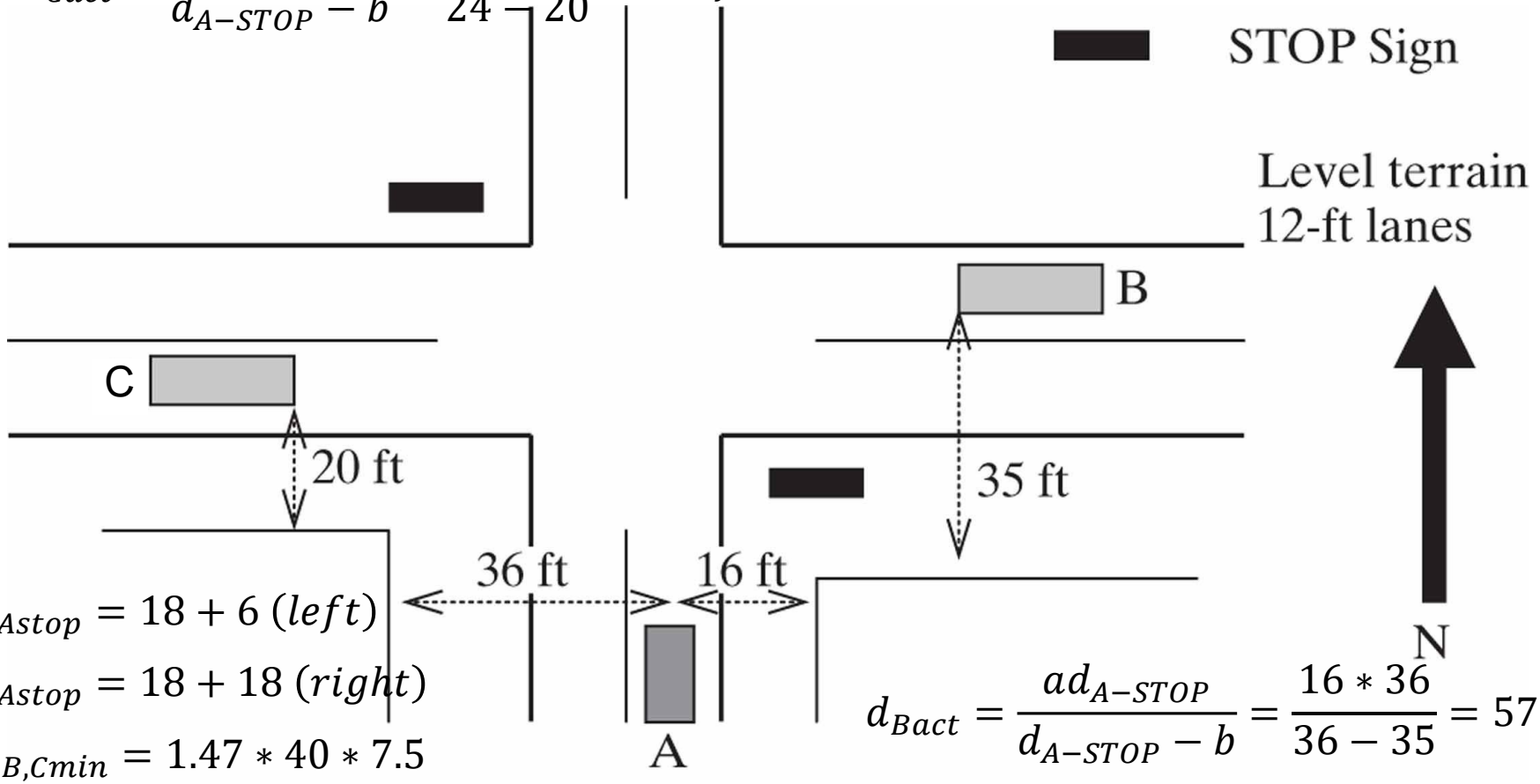
Example



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Example

$$d_{Cact} = \frac{ad_{A-STOP}}{d_{A-STOP} - b} = \frac{36 * 24}{24 - 20} = 216ft$$



$$d_{Astop} = 18 + 6\ (left)$$

$$d_{Astop} = 18 + 18\ (right)$$

$$d_{B,Cmin} = 1.47 * 40 * 7.5$$

=441ft

$$d_{Bact} = \frac{ad_{A-STOP}}{d_{A-STOP} - b} = \frac{16 * 36}{36 - 35} = 576ft$$

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Warrants for multiway stop signs (Section 2B.07 Multi-Way Stop Applications)

- Guidance:
- The decision to install multi-way stop control *should* be based on an engineering study.
- The following criteria *should* be considered in the engineering study for a multi-way STOP sign installation:

A. Where traffic control signals are justified, the multi-way stop is an interim measure that can be installed quickly to control traffic while arrangements are being made for the installation of the traffic control signal.

B. Five or more reported **crashes** in a 12-month period that are susceptible to correction by a multi-way stop installation. Such crashes include right-turn and left-turn collisions as well as right-angle collisions.

C. **Minimum volumes**:

1. The vehicular volume entering the intersection from the major street approaches (total of both approaches) averages at least 300 vehicles per hour for any 8 hours of an average day; and
2. The combined vehicular, pedestrian, and bicycle volume entering the intersection from the minor street approaches (total of both approaches) averages at least 200 units per hour for the same 8 hours, with an average delay to minor-street vehicular traffic of at least 30 seconds per vehicle during the highest hour; but
3. If the 85th-percentile approach speed of the major-street traffic exceeds 40 mph, the minimum vehicular volume warrants are 70 percent of the values provided in Items 1 and 2.

D. Where no single criterion is satisfied, but where Criteria B, C.1, and C.2 are all satisfied to 80 percent of the minimum values. Criterion C.3 is excluded from this condition.

Warrants for yield signs (Section 2B.09 YIELD Sign Applications)

- Option:
- YIELD signs *may* be installed:

A. On the approaches to a through street or highway where conditions are such that a full stop is not always required.

B. At the second crossroad of a divided highway, where the **median width at the intersection is 30 feet or greater**. In this case, a STOP or YIELD sign may be installed at the entrance to the first roadway of a divided highway, and a YIELD sign may be installed at the entrance to the second roadway.

C. For a **channelized turn lane** that is separated from the adjacent travel lanes by an island, even if the adjacent lanes at the intersection are controlled by a highway traffic control signal or by a STOP sign.

D. At an intersection where a special problem exists and where engineering judgment indicates the problem to be susceptible to correction by the use of the YIELD sign.

E. Facing the entering roadway for a merge-type movement if engineering judgment indicates that control is needed because acceleration geometry and/or sight distance is not adequate for merging traffic operation.

Warrants for yield signs (Section 2B.09 YIELD Sign Applications)

- Standard:
- A YIELD (R1-2) sign *shall* be used to assign right-of-way at the entrance to a **roundabout**. YIELD signs at roundabouts shall be used to control the approach roadways and shall not be used to control the circulatory roadway.
- Other than for all of the approaches to a roundabout, YIELD signs *shall not* be placed on all of the approaches to an intersection.

What is a roundabout?

- A roundabout is a one-way circular intersection without stop signs or traffic signals.
- Unlike the traffic circles of the past, modern roundabouts are typically smaller, have slower speeds and require entering vehicles to yield to circulating traffic.



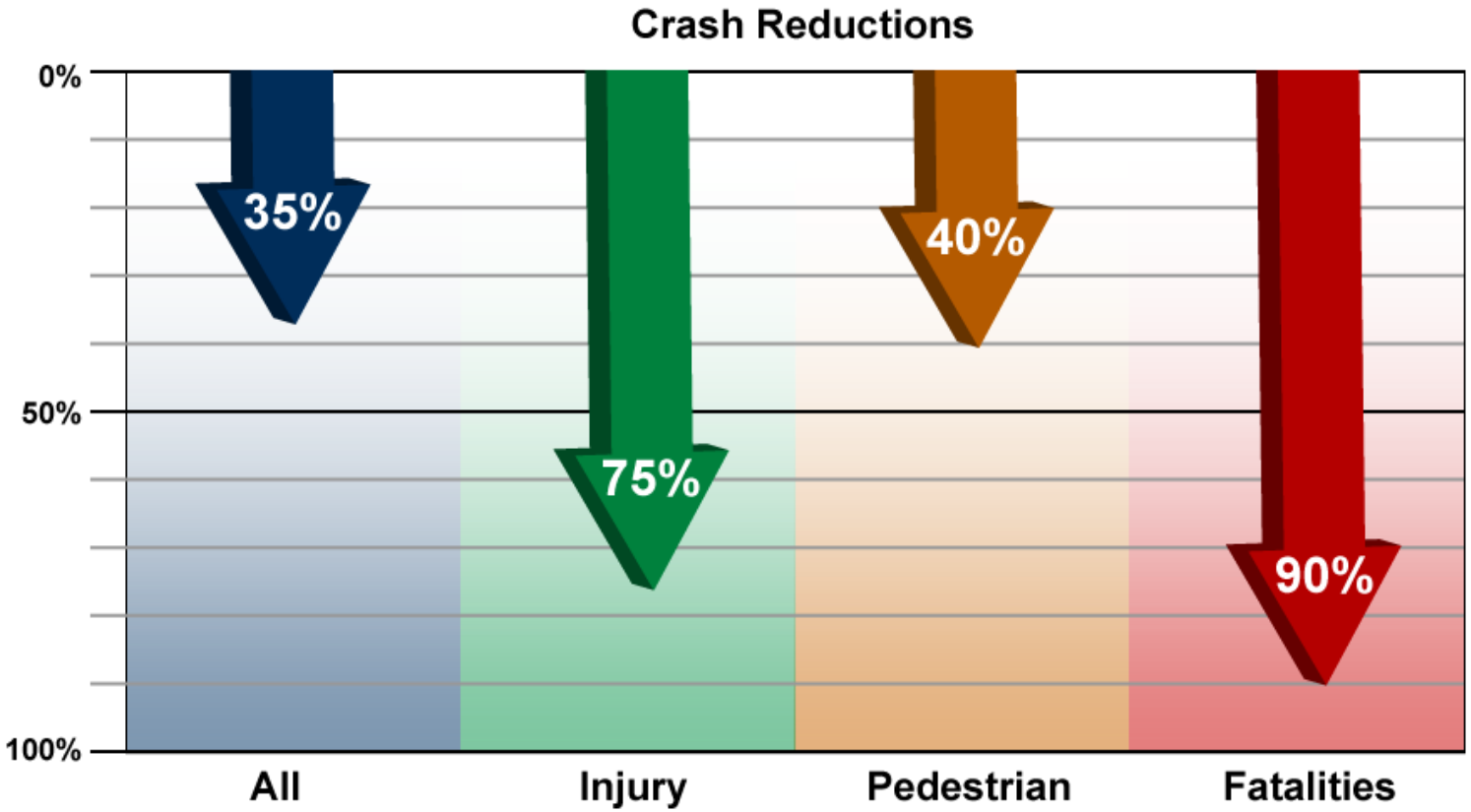
(source: scdot.org)

Why install a roundabout?

- Roundabouts are replacing traditional intersections because they work well. In the right location, roundabouts keep you safer and reduce your travel time through an intersection.

(source: scdot.org)

Roundabout enhanced safety



(source: scdot.org)

Other benefits of roundabout

- Enhanced operation
  - Drivers will experience a reduction in delays and queues at roundabouts, especially during peak travel times.
- Enhanced air quality
  - Fewer stops, starts, shorter queues, and delays, result in a reduction in vehicular emissions and fuel consumption.
- Enhanced aesthetics
  - Landscaping in the central island improves the appearance.

(source: scdot.org)

Level III control: traffic control signals

- Traffic signals assign right-of-way to specific movements, and thus, it can reduce the number and nature of intersection conflicts.
- Pedestrian-vehicle and bicycle-vehicle conflicts remain between legal movements, and driver vigilance and judgment are still required to avoid accidents.
- Traffic signals are **costly to install and maintain** and they introduce a **fixed source of delay** into the system. Therefore, it is important that they are not overused; they should be installed only where no other solution for form of control would be effective in assuring safety and efficiency at the intersection.

Warrants for traffic signals

(Section 4C.01 Studies and Factors for Justifying Traffic Control Signals)

- Standard:
- An engineering study of traffic conditions, pedestrian characteristics, and physical characteristics of the location *shall* be performed to determine whether installation of a traffic control signal is justified at a particular location.
- The investigation of the need for a traffic control signal *shall* include an analysis of factors related to the existing operation and safety at the study location and the potential to improve these conditions, and the applicable factors contained in the following traffic signal warrants:
  - Warrant 1, Eight-Hour Vehicular Volume
  - Warrant 2, Four-Hour Vehicular Volume
  - Warrant 3, Peak Hour
  - Warrant 4, Pedestrian Volume
  - Warrant 5, School Crossing
  - Warrant 6, Coordinated Signal System
  - Warrant 7, Crash Experience
  - Warrant 8, Roadway Network
  - Warrant 9, Intersection Near a Grade Crossing
- The satisfaction of a traffic signal warrant or warrants *shall* not in itself require the installation of a traffic control signal.

Warrants for traffic signals

(Section 4C.01 Studies and Factors for Justifying Traffic Control Signals)

Option:

- **Engineering study data** may include the following:

A. The number of vehicles entering the intersection in each hour from each approach during 12 hours of an average day. It is desirable that the hours selected contain the greatest percentage of the 24-hour traffic volume.

B. Vehicular volumes for each traffic movement from each approach, classified by vehicle type (heavy trucks, passenger cars and light trucks, public-transit vehicles, and, in some locations, bicycles), during each 15-minute period of the 2 hours in the morning and 2 hours in the afternoon during which total traffic entering the intersection is greatest.

C. Pedestrian volume counts on each crosswalk during the same periods as the vehicular counts in Item B and during hours of highest pedestrian volume. Where young, elderly, and/or persons with physical or visual disabilities need special consideration, the pedestrians and their crossing times may be classified by general observation.

D. Information about nearby facilities and activity centers that serve the young, elderly, and/or persons with disabilities, including requests from persons with disabilities for accessible crossing improvements at the location under study. These persons might not be adequately reflected in the pedestrian volume count if the absence of a signal restrains their mobility.

E. The posted or statutory speed limit or the 85th-percentile speed on the uncontrolled approaches to the location.

F. A condition diagram showing details of the physical layout, including such features as intersection geometrics, channelization, grades, sight-distance restrictions, transit stops and routes, parking conditions, pavement markings, roadway lighting, driveways, nearby railroad crossings, distance to nearest traffic control signals, utility poles and fixtures, and adjacent land use.

G. A collision diagram showing crash experience by type, location, direction of movement, severity, weather, time of day, date, and day of week for at least 1 year.

Warrant 1, Eight-Hour Vehicular Volume (Section 4C.02)

- Standard:
- The need for a traffic control signal *shall* be considered if an engineering study finds that one of the following conditions exist for each of any 8 hours of an average day:

A. The vehicles per hour given in both of the 100 percent columns of Condition A in Table 4C-1 exist on the major-street and the higher-volume minor-street approaches, respectively, to the intersection; or

B. The vehicles per hour given in both of the 100 percent columns of Condition B in Table 4C-1 exist on the major-street and the higher-volume minor-street approaches, respectively, to the intersection.

- In applying each condition the major-street and minor-street volumes shall be for the same 8 hours. On the minor street, the higher volume shall not be required to be on the same approach during each of these 8 hours.

Table 4C-1. Warrant 1, Eight-Hour Vehicular Volume

Condition A—Minimum Vehicular Volume

Number of lanes for moving traffic on each approach		Vehicles per hour on major street (total of both approaches)				Vehicles per hour on higher-volume minor-street approach (one direction only)			
Major Street	Minor Street	100%[a]	80%[b]	70%[c]	56%[d]	100%[a]	80%[b]	70%[c]	56%[d]
1	1	500	400	350	280	150	120	105	84
2 or more	1	600	480	420	336	150	120	105	84
2 or more	2 or more	600	480	420	336	200	160	140	112
1	2 or more	500	400	350	280	200	160	140	112

Condition B—Interruption of Continuous Traffic

Number of lanes for moving traffic on each approach		Vehicles per hour on major street (total of both approaches)				Vehicles per hour on higher-volume minor-street approach (one direction only)			
Major Street	Minor Street	100%[a]	80%[b]	70%[c]	56%[d]	100%[a]	80%[b]	70%[c]	56%[d]
1	1	750	600	525	420	75	60	53	42
2 or more	1	900	720	630	504	75	60	53	42
2 or more	2 or more	900	720	630	504	100	80	70	56
1	2 or more	750	600	525	420	100	80	70	56

[a] Basic minimum hourly volume

[b] Used for combination of Conditions A and B after adequate trial of other remedial measures

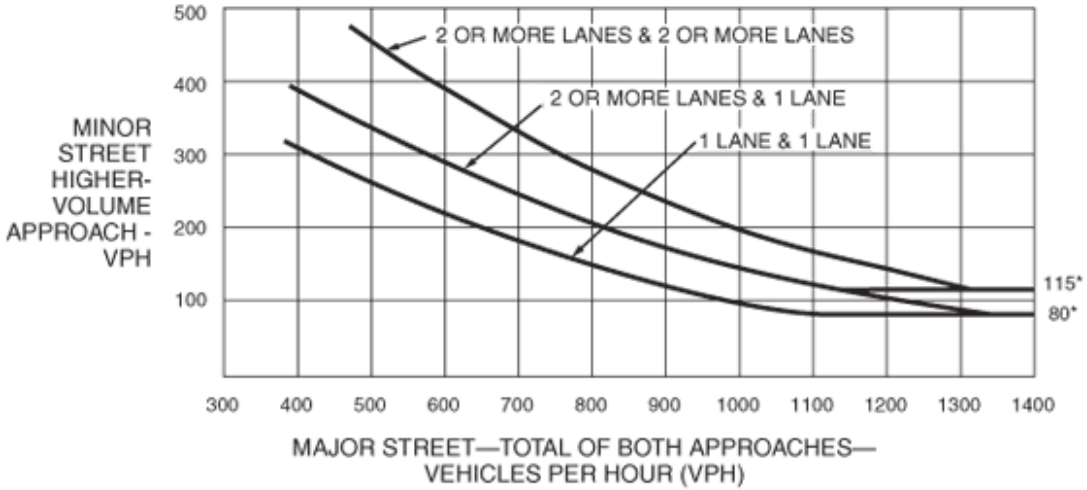
[c] May be used when the major-street speed exceeds 40 mph or in an isolated community with a population of less than 10,000

[d] May be used for combination of Conditions A and B after adequate trial of other remedial measures when the major-street speed exceeds 40 mph or in an isolated community with a population of less than 10,000

Warrant 2, Four-Hour Vehicular Volume (Section 4C.03)

- Standard:
- The need for a traffic control signal shall be considered if an engineering study finds that, for each of any 4 hours of an average day, the plotted points representing the vehicles per hour on the major street (total of both approaches) and the corresponding vehicles per hour on the higher-volume minor-street approach (one direction only) all fall above the applicable curve in Figure 4C-1 for the existing combination of approach lanes. On the minor street, the higher volume shall not be required to be on the same approach during each of these 4 hours.

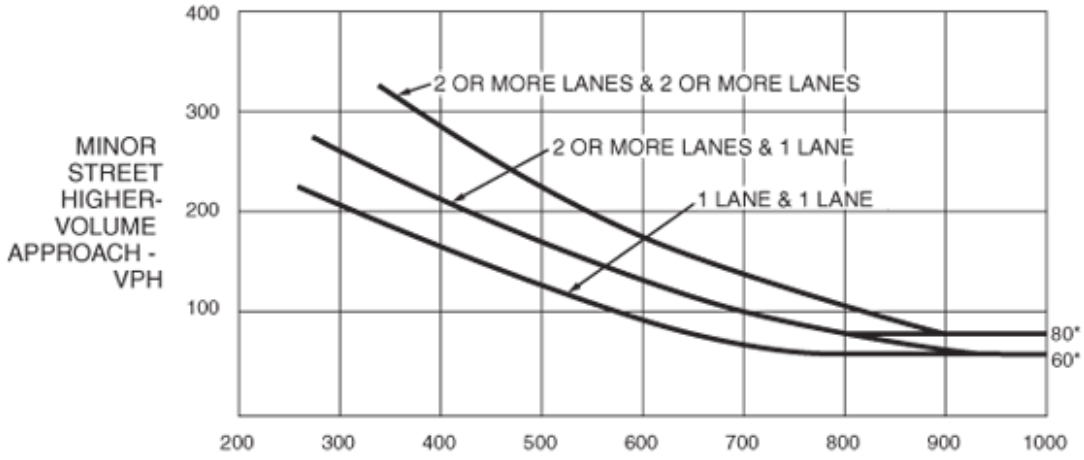
Figure 4C-1. Warrant 2, Four-Hour Vehicular Volume



*Note: 115 vph applies as the lower threshold volume for a minor-street approach with two or more lanes and 80 vph applies as the lower threshold volume for a minor-street approach with one lane.

Figure 4C-2. Warrant 2, Four-Hour Vehicular Volume (70% Factor)

(COMMUNITY LESS THAN 10,000 POPULATION OR ABOVE 40 MPH ON MAJOR STREET)



*Note: 80 vph applies as the lower threshold volume for a minor-street approach with two or more lanes and 60 vph applies as the lower threshold volume for a minor-street approach with one lane.

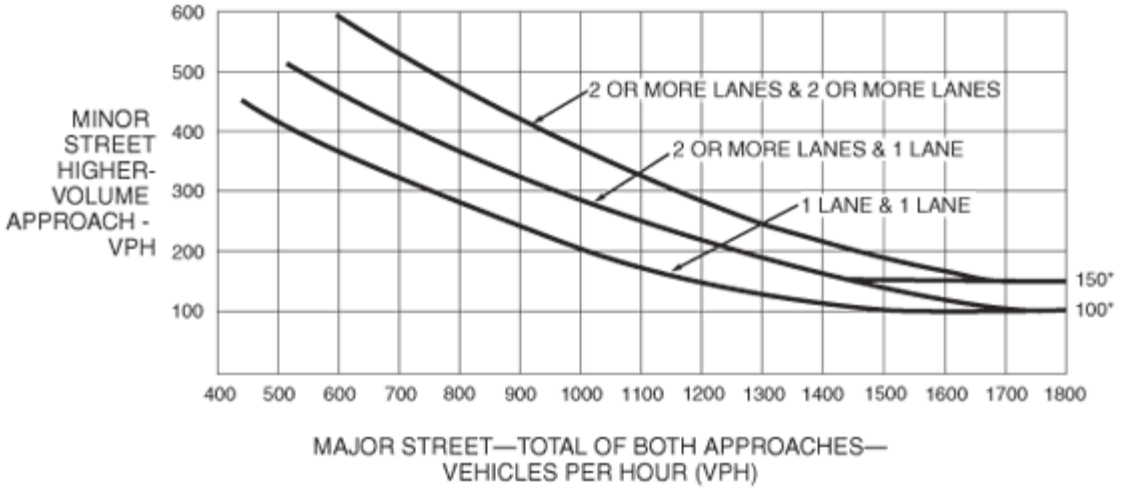
Warrant 3, Peak Hour (Section 4C.04)

- Standard:
- This signal warrant shall be applied only in unusual cases, such as office complexes, manufacturing plants, industrial complexes, or high-occupancy vehicle facilities that attract or discharge large numbers of vehicles over a short time.
- The need for a traffic control signal shall be considered if an engineering study finds that the **criteria in either of the following two categories are met:**

A. If all three of the following conditions exist for the same 1 hour (any four consecutive 15-minute periods) of an average day:
   1. The total stopped time delay experienced by the traffic on one minor-street approach (one direction only) controlled by a STOP sign equals or exceeds: 4 vehicle-hours for a one-lane approach or 5 vehicle-hours for a two-lane approach; and
   2. The volume on the same minor-street approach (one direction only) equals or exceeds 100 vehicles per hour for one moving lane of traffic or 150 vehicles per hour for two moving lanes; and
   3. The total entering volume serviced during the hour equals or exceeds 650 vehicles per hour for intersections with three approaches or 800 vehicles per hour for intersections with four or more approaches.

B. The plotted point representing the vehicles per hour on the major street (total of both approaches) and the corresponding vehicles per hour on the higher-volume minor-street approach (one direction only) for 1 hour (any four consecutive 15-minute periods) of an average day falls above the applicable curve in Figure 4C-3 for the existing combination of approach lanes.

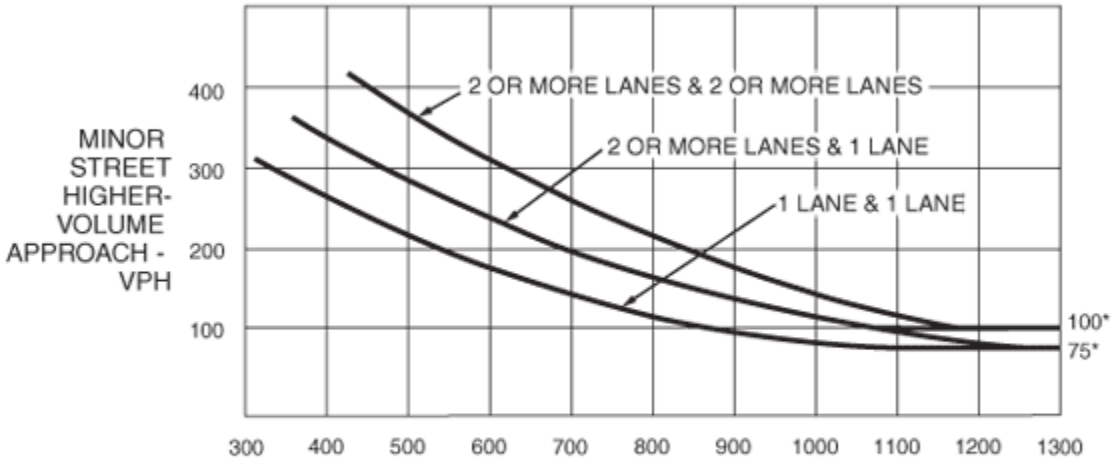
Figure 4C-3. Warrant 3, Peak Hour



*Note: 150 vph applies as the lower threshold volume for a minor-street approach with two or more lanes and 100 vph applies as the lower threshold volume for a minor-street approach with one lane.

Figure 4C-4. Warrant 3, Peak Hour (70% Factor)

(COMMUNITY LESS THAN 10,000 POPULATION OR ABOVE 40 MPH ON MAJOR STREET)



*Note: 100 vph applies as the lower threshold volume for a minor-street approach with two or more lanes and 75 vph applies as the lower threshold volume for a minor-street approach with one lane.

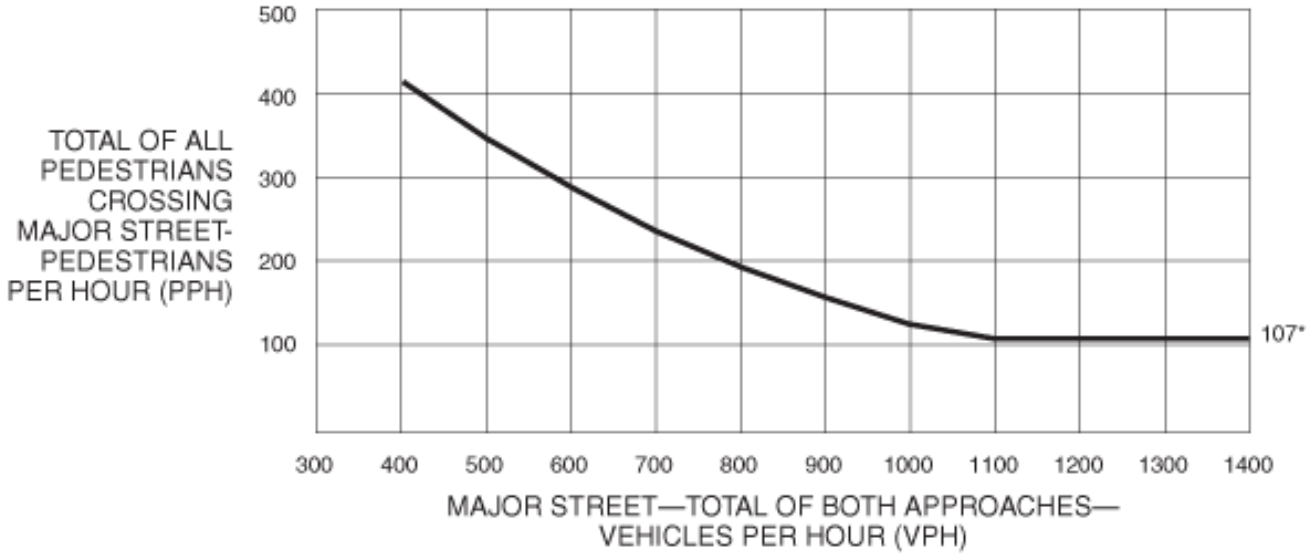
Warrant 4, Pedestrian Volume (Section 4C.05)

- Standard:
- The need for a traffic control signal at an intersection or midblock crossing shall be considered if an engineering study finds that **one of the following criteria is met**:

A. For each of any 4 hours of an average day, the plotted points representing the vehicles per hour on the major street (total of both approaches) and the corresponding pedestrians per hour crossing the major street (total of all crossings) all fall above the curve in Figure 4C-5; or

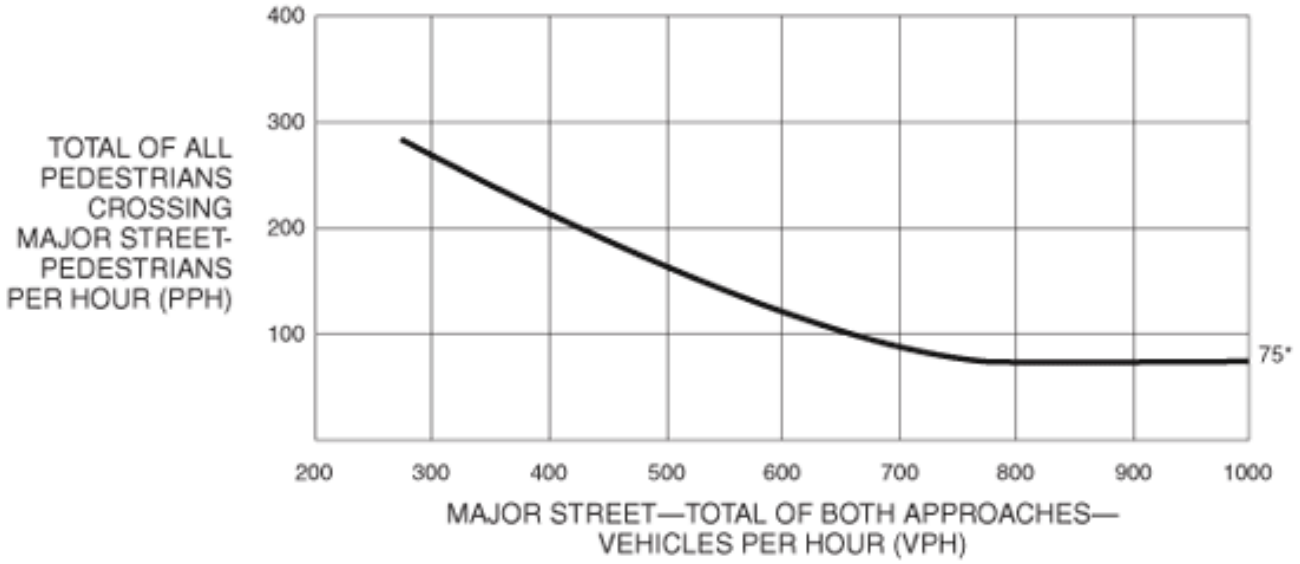
B. For 1 hour (any four consecutive 15-minute periods) of an average day, the plotted point representing the vehicles per hour on the major street (total of both approaches) and the corresponding pedestrians per hour crossing the major street (total of all crossings) falls above the curve in Figure 4C-7.

Figure 4C-5. Warrant 4, Pedestrian Four-Hour Volume



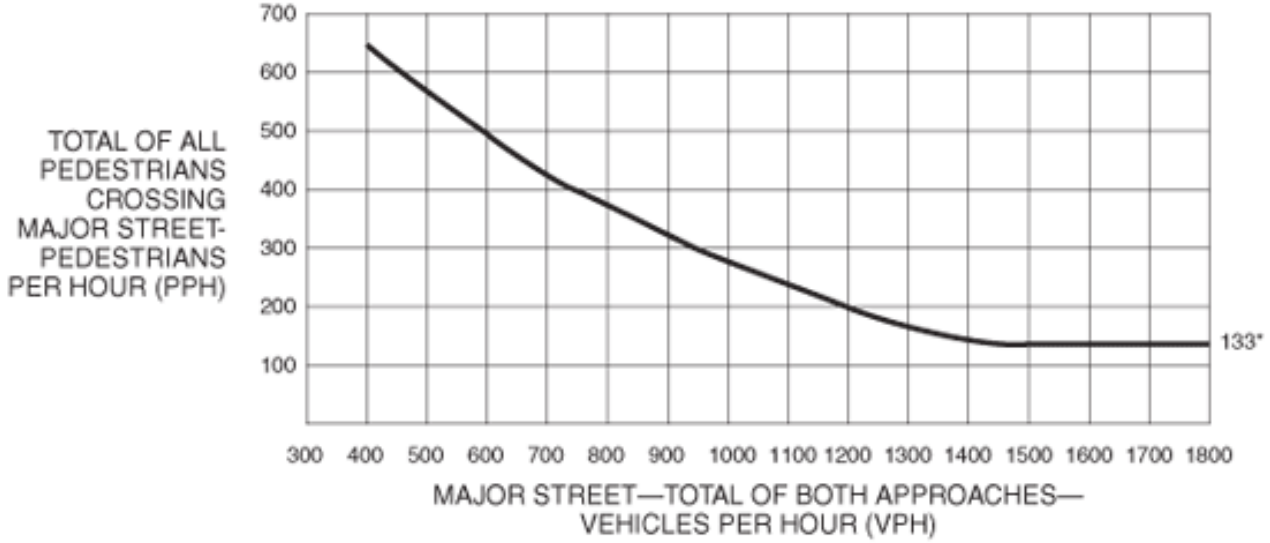
*Note: 107 pph applies as the lower threshold volume.

Figure 4C-6. Warrant 4, Pedestrian Four-Hour Volume (70% Factor)



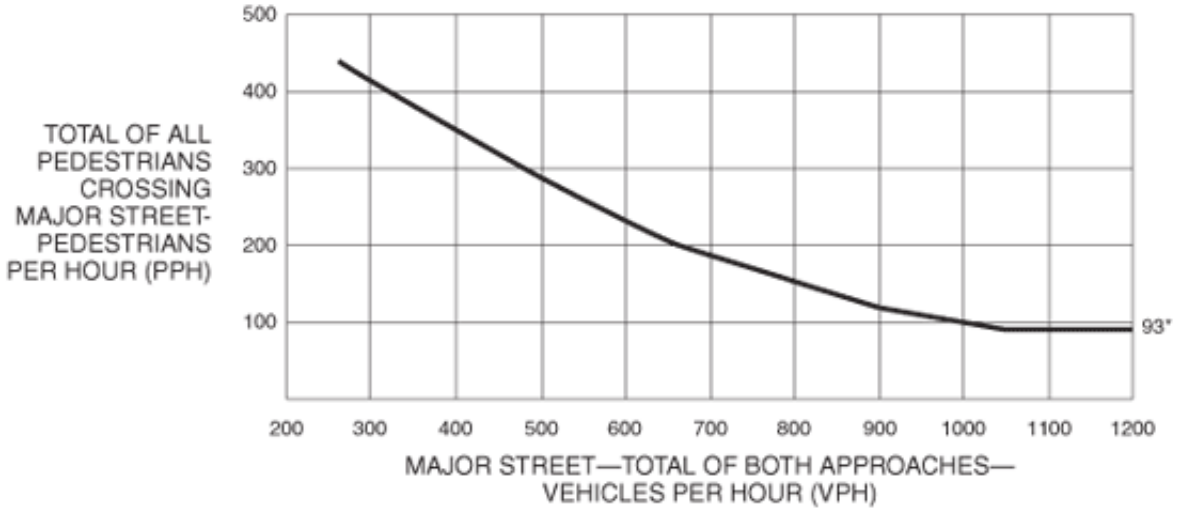
*Note: 75 pph applies as the lower threshold volume.

Figure 4C-7. Warrant 4, Pedestrian Peak Hour



*Note: 133 pph applies as the lower threshold volume.

Figure 4C-8. Warrant 4, Pedestrian Peak Hour (70% Factor)



*Note: 93 pph applies as the lower threshold volume.

Warrant 5, School Crossing (Section 4C.06)

- Standard:
- The need for a traffic control signal *shall* be considered when an engineering study of the frequency and adequacy of gaps in the vehicular traffic stream as related to the number and size of groups of schoolchildren at an established school crossing across the major street shows that the number of adequate gaps in the traffic stream during the period when the schoolchildren are using the crossing is less than the number of minutes in the same period (see Section 7A.03) and there are a minimum of 20 schoolchildren during the highest crossing hour.
- Before a decision is made to install a traffic control signal, consideration shall be given to the implementation of other remedial measures, such as warning signs and flashers, school speed zones, school crossing guards, or a grade-separated crossing.
- The School Crossing signal warrant shall not be applied at locations where the distance to the nearest traffic control signal along the major street is less than 300 feet, unless the proposed traffic control signal will not restrict the progressive movement of traffic.

Warrant 6, Coordinated Signal System (Section 4C.07)

- Standard:
- The need for a traffic control signal shall be considered if an engineering study finds that **one of the following criteria is met**:

A. **On a one-way street** or a street that has traffic predominantly in one direction, the adjacent traffic control signals are so far apart that they do not provide the necessary degree of vehicular platooning.

B. **On a two-way street**, adjacent traffic control signals do not provide the necessary degree of platooning and the proposed and adjacent traffic control signals will collectively provide a progressive operation.

Warrant 7, Crash Experience (Section 4C.08)

Standard:

- The need for a traffic control signal shall be considered if an engineering study finds that **all of the following criteria are met**:

A. Adequate trial of alternatives with satisfactory observance and enforcement has failed to reduce the crash frequency; and

B. Five or more reported crashes, of types susceptible to correction by a traffic control signal, have occurred within a 12-month period, each crash involving personal injury or property damage apparently exceeding the applicable requirements for a reportable crash; and

C. For each of any 8 hours of an average day, the vehicles per hour (vph) given in both of the 80 percent columns of Condition A in Table 4C-1 (see Section 4C.02), or the vph in both of the 80 percent columns of Condition B in Table 4C-1 exists on the major-street and the higher-volume minor-street approach, respectively, to the intersection, or the volume of pedestrian traffic is not less than 80 percent of the requirements specified in the Pedestrian Volume warrant. These major-street and minor-street volumes shall be for the same 8 hours. On the minor street, the higher volume shall not be required to be on the same approach during each of the 8 hours.

Warrant 8, Roadway Network (Section 4C.09)

- Standard:
- The need for a traffic control signal shall be considered if an engineering study finds that the common intersection of two or more major routes meets **one or both of the following criteria**:

A. The intersection has a total existing, or immediately projected, entering volume of at least 1,000 vehicles per hour during the peak hour of a typical weekday and has 5-year projected traffic volumes, based on an engineering study, that meet one or more of Warrants 1, 2, and 3 during an average weekday; or

B. The intersection has a total existing or immediately projected entering volume of at least 1,000 vehicles per hour for each of any 5 hours of a non-normal business day (Saturday or Sunday).

- A major route as used in this signal warrant shall have **at least one of the following characteristics**:

A. It is part of the street or highway system that serves as the principal roadway network for through traffic flow.

B. It includes rural or suburban highways outside, entering, or traversing a city.

C. It appears as a major route on an official plan, such as a major street plan in an urban area traffic and transportation study.

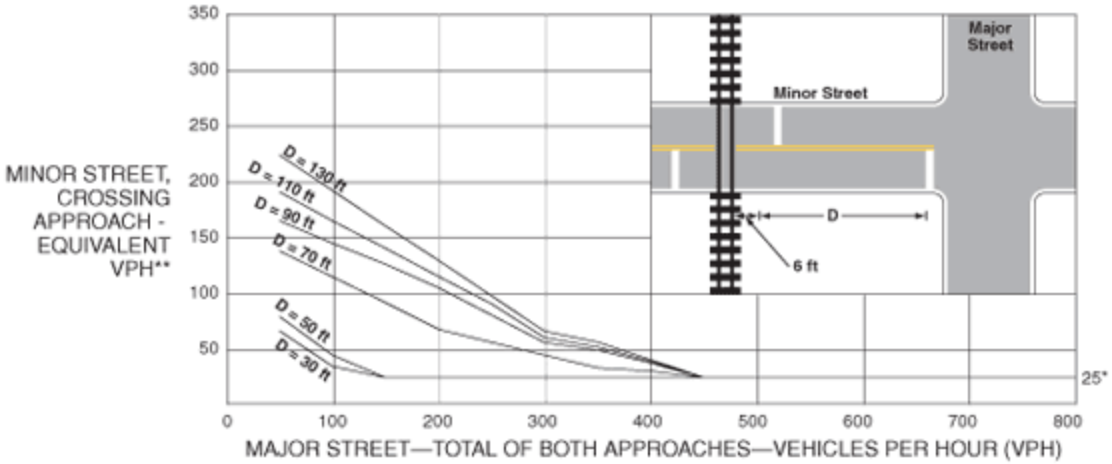
Warrant 9, Intersection Near a Grade Crossing (Section 4C.09)

- Standard:
- The need for a traffic control signal shall be considered if an engineering study finds that **both of the following criteria are met**:

A. A grade crossing exists on an approach controlled by a STOP or YIELD sign and the center of the track nearest to the intersection is within 140 feet of the stop line or yield line on the approach; and

B. During the highest traffic volume hour during which rail traffic uses the crossing, the plotted point representing the vehicles per hour on the major street (total of both approaches) and the corresponding vehicles per hour on the minor-street approach that crosses the track (one direction only, approaching the intersection) falls above the applicable curve in Figure 4C-9 or 4C-10 for the existing combination of approach lanes over the track and the distance D, which is the clear storage distance as defined in Section 1A.13.

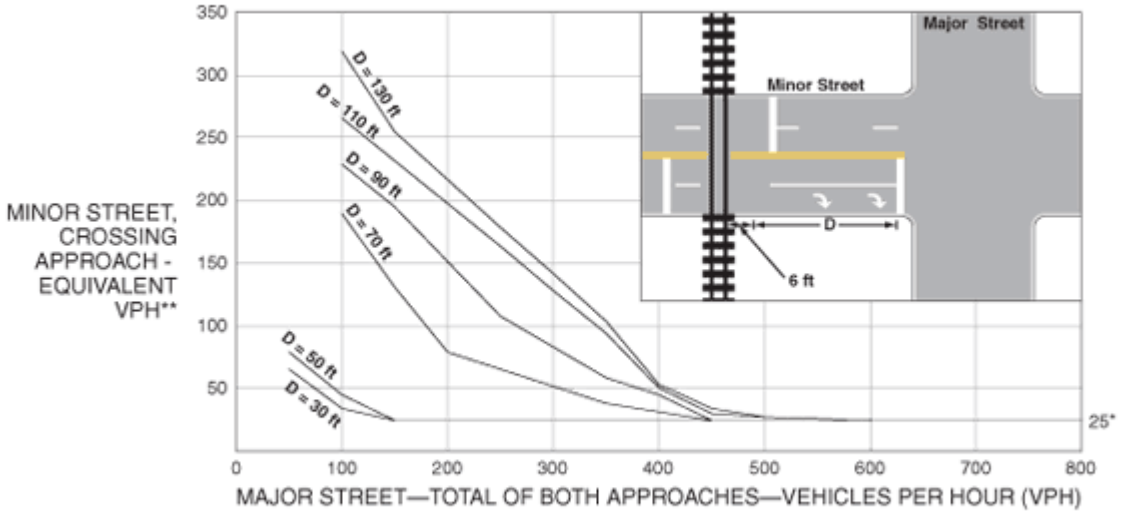
Figure 4C-9. Warrant 9, Intersection Near a Grade Crossing (One Approach Lane at the Track Crossing)



* 25 vph applies as the lower threshold volume

** VPH after applying the adjustment factors in Tables 4C-2, 4C-3, and/or 4C-4, if appropriate

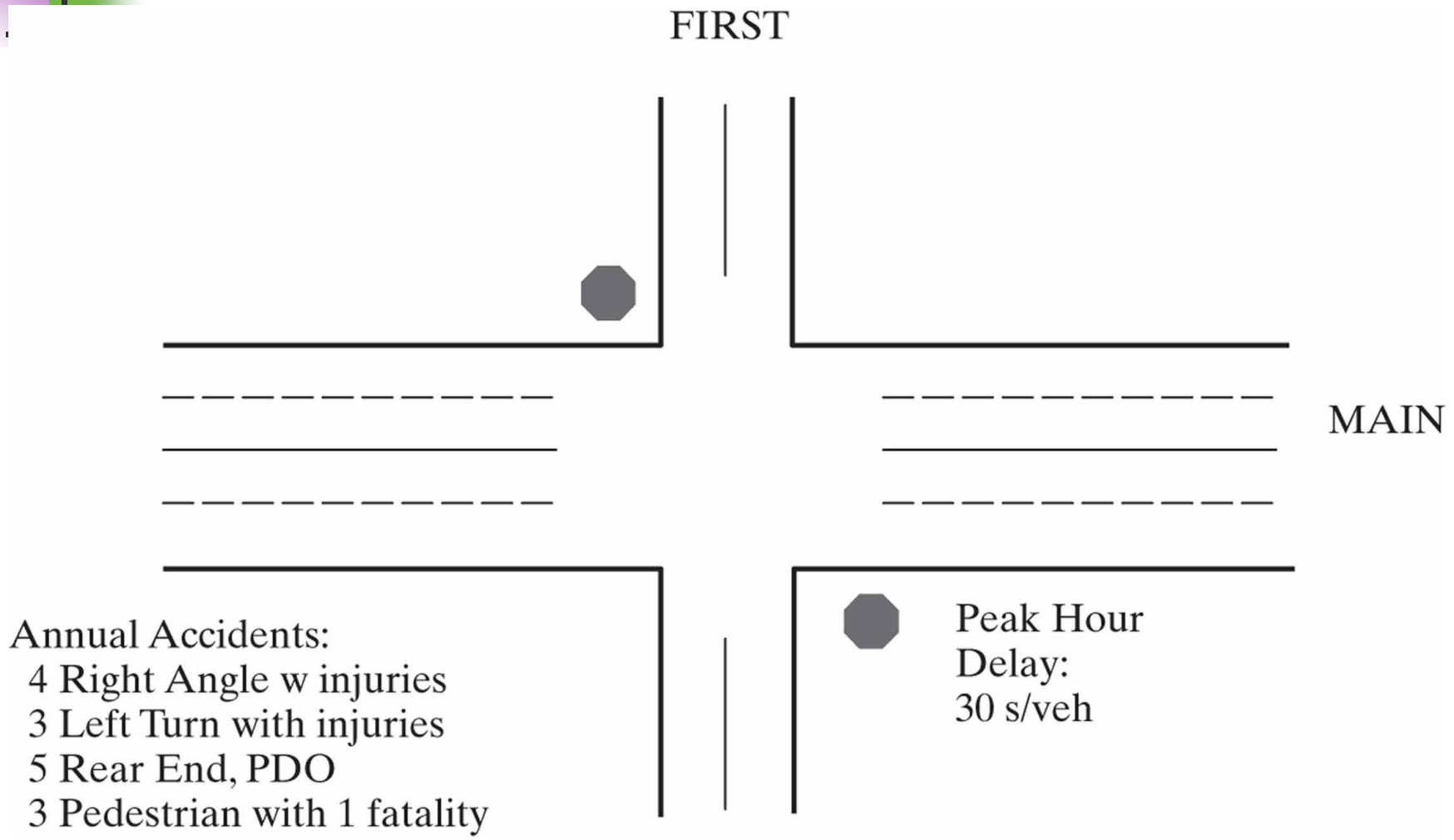
Figure 4C-10. Warrant 9, Intersection Near a Grade Crossing (Two or More Approach Lanes at the Track Crossing)



* 25 vph applies as the lower threshold volume

** VPH after applying the adjustment factors in Tables 4C-2, 4C-3, and/or 4C-4, if appropriate

Example application of signal warrants



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Data for example problem

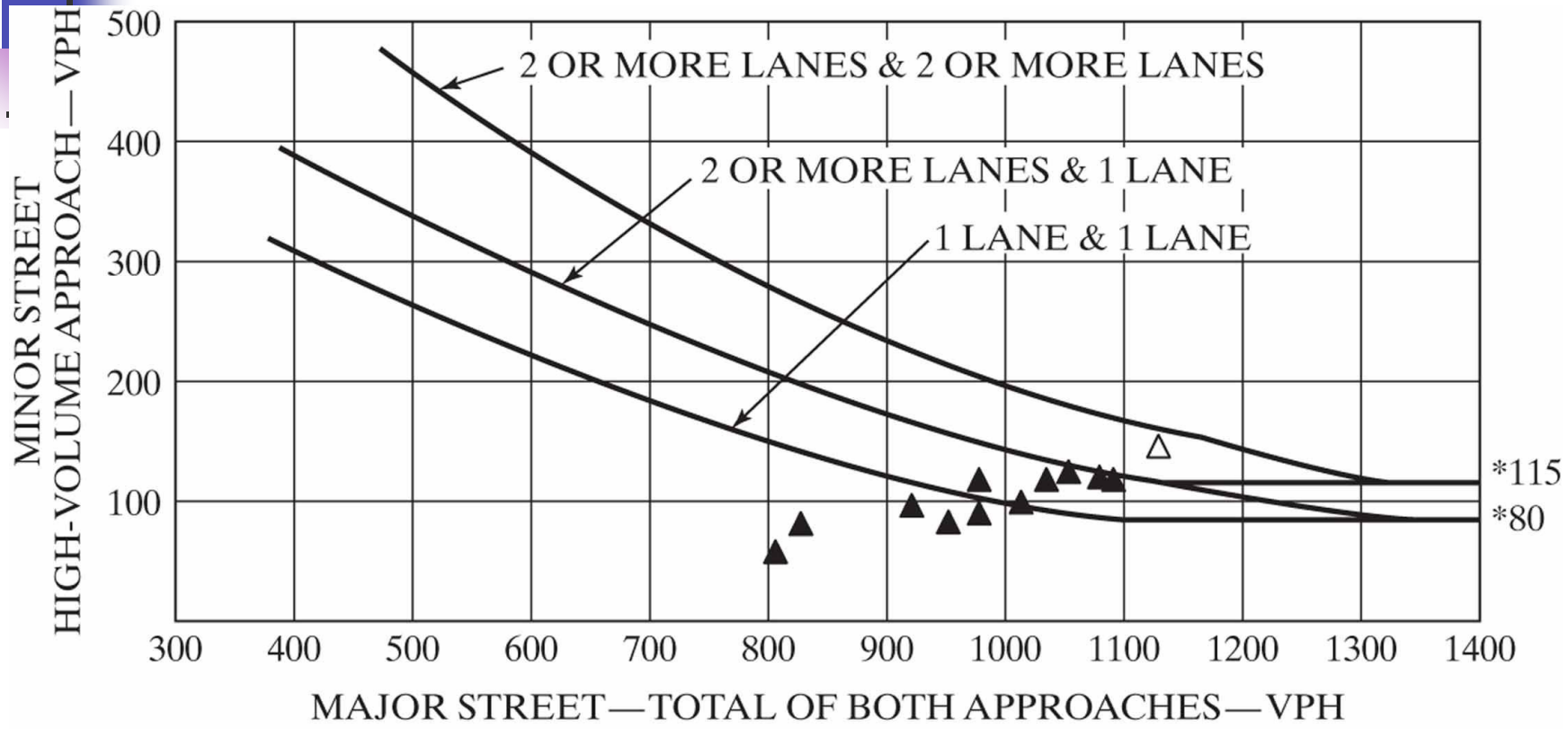
Time	Main Street Volume (veh/h)			First Ave Volume (veh/h)			Ped Volume (ped/h)
	EB	WB	TOT	NB	SB	High Vol	Xing Main
11 AM–12	400	425	**825**	75	80	**80**	115
12–1 PM	450	465	**915**	85	85	**85**	120
1–2 PM	485	500	**985**	90	100	**100**	125
2–3 PM	525	525	**1,050**	110	115	**115**	130
3–4 PM	515	525	**1,040**	100	95	**100**	135
4–5 PM	540	550	**1,090**	90	100	**100**	140
5–6 PM	550	580	**1,130**	110	125	**125**	120
6–7 PM	545	525	**1,070**	96	103	**103**	108
7–8 PM	505	506	**1,011**	90	95	**95**	100
8–9 PM	485	490	**975**	85	75	**85**	90
9–10 PM	475	475	**950**	75	60	**75**	50
10–11 PM	400	410	**810**	50	55	**55**	25

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Warrants Summary

- Warrant 1: met at 100% level.
  - Condition A is not met
  - Condition B is met (between 12 & 10 PM)
- Warrant 2: not met. Only one of the 12 hours where volume is above the decision curve.

Example Application of Warrant 2



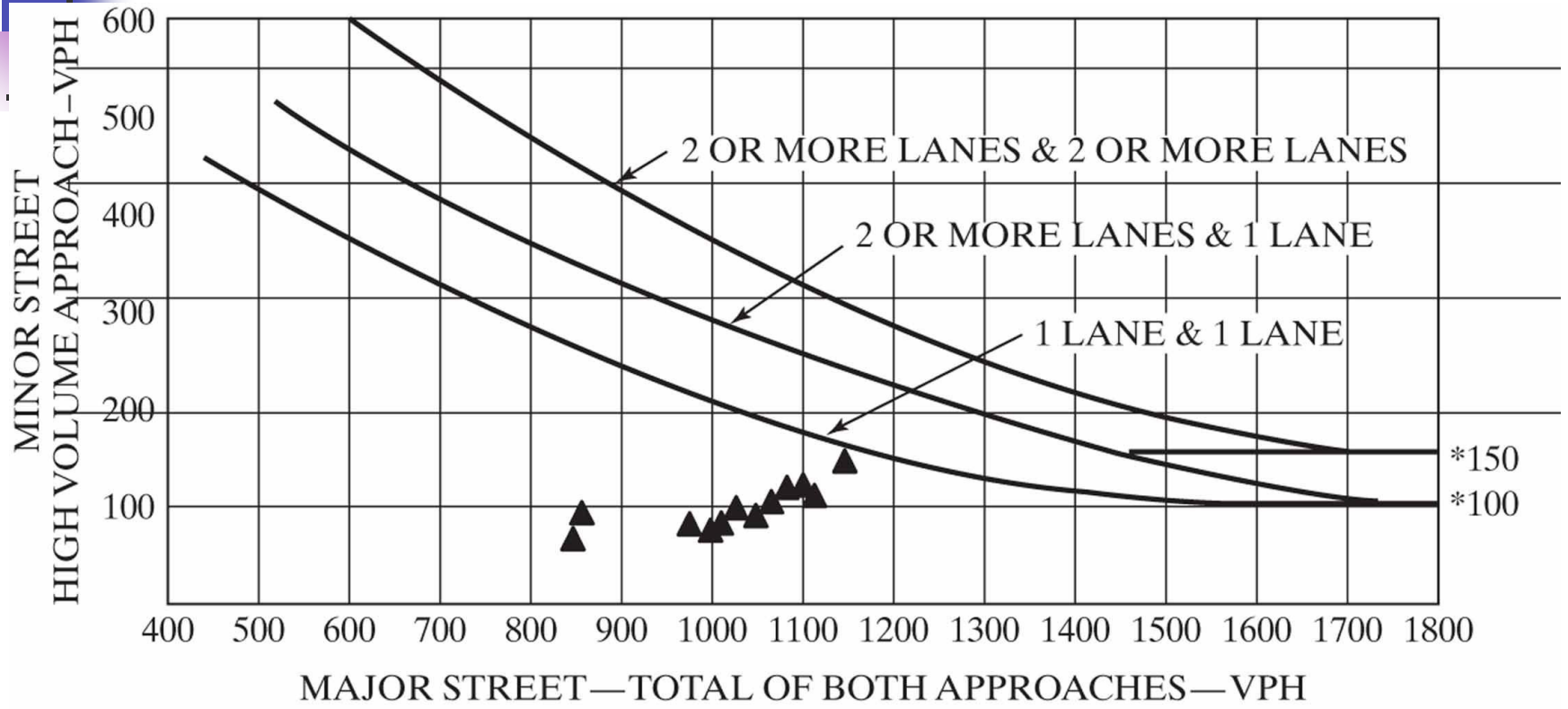
*Note: 115 vph applies as the lower threshold volume for a minor street approach with two or more lanes and 80 vph applies as the lower threshold volume for a minor street approach with one lane.

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Warrants Summary

- Warrant 3: not met.
  - None of the 12 hours had a volume above the decision curve.
  - The total vehicle delay during the peak hour is 125 veh * 30 sec = 3,750 veh-sec or 1.04 veh-hr < 4 veh-hr

Example Application of Warrant 3



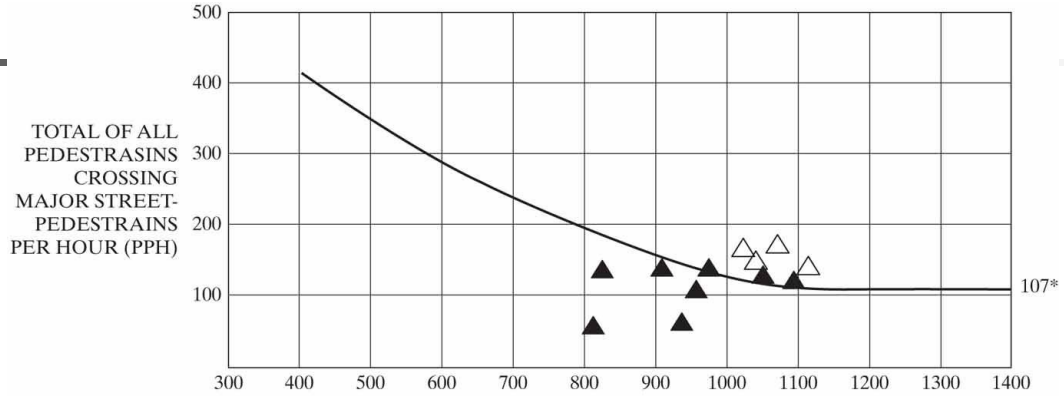
*Note: 150 vph applies as the lower threshold volume for a minor street approach with two or more lanes and 100 vph applies as the lower threshold volume for a minor street approach with one lane.

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Warrants Summary

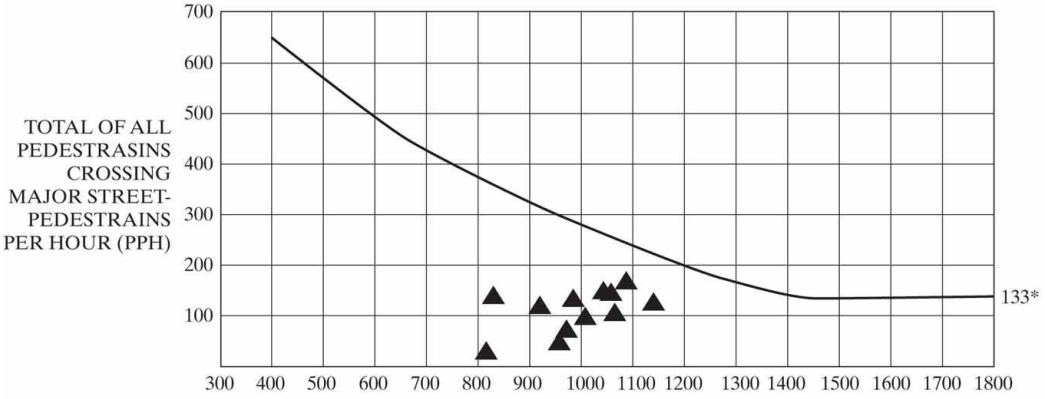
- Warrant 4: is met
  - 4-hour pedestrian warrant is met
  - Peak hour pedestrian warrant is not met

Example Application of Warrant 4



*Note: 107 pph applies as the lower thershold volume.

(a) Four-Hour Pedestrian Warrant



*Note: 133 pph applies as the lower thershold colume.

(b) Peak-Hour Pedestrian Warrant

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Warrants Summary

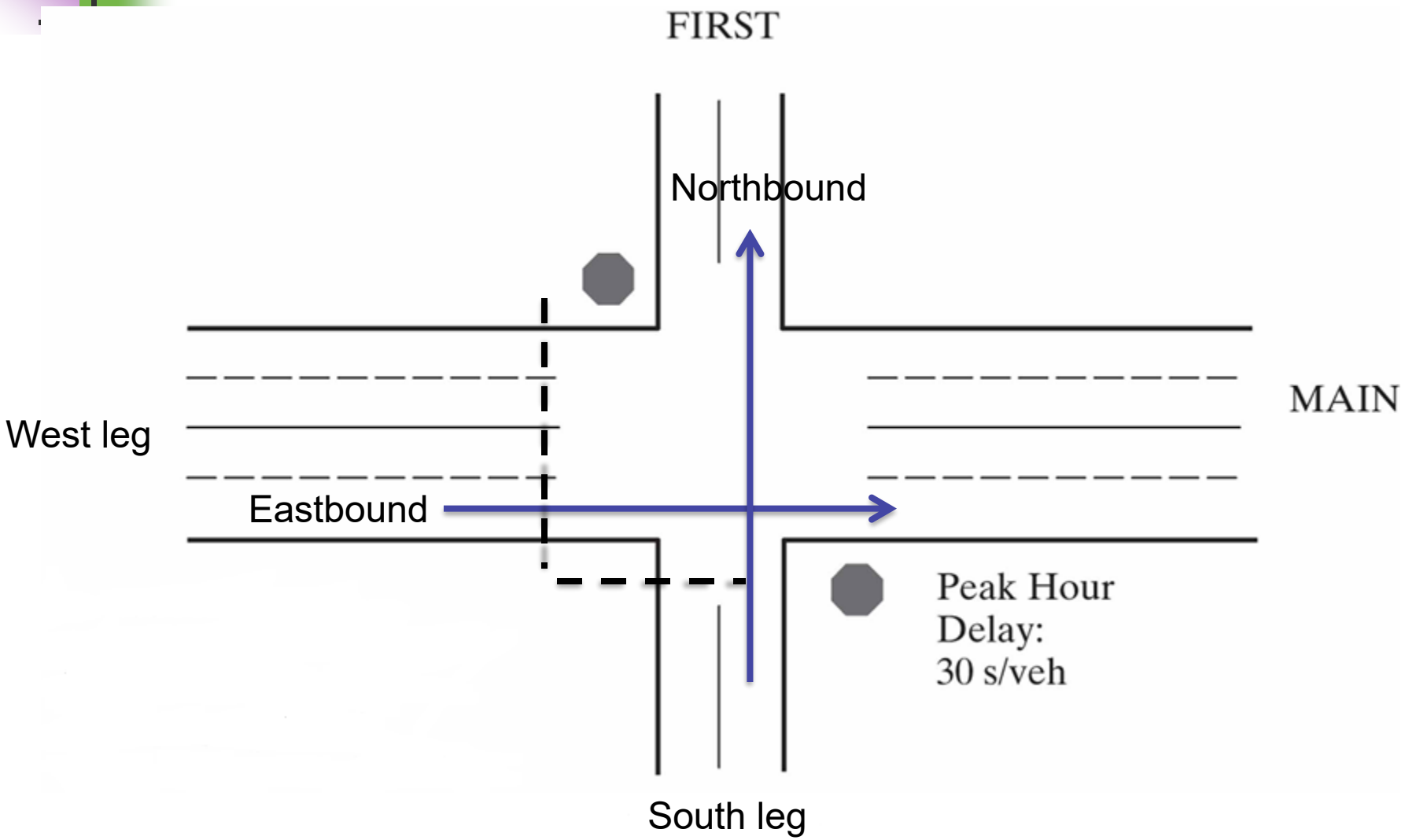
- Warrant 5: N/A
- Warrant 6: N/A
- Warrant 7: is met
  - All measures have been tried.
  - Signal control can correct the 4 right-angle, 2 left-turn, and 3 pedestrian related crashes.
  - Warrant 1B is met.

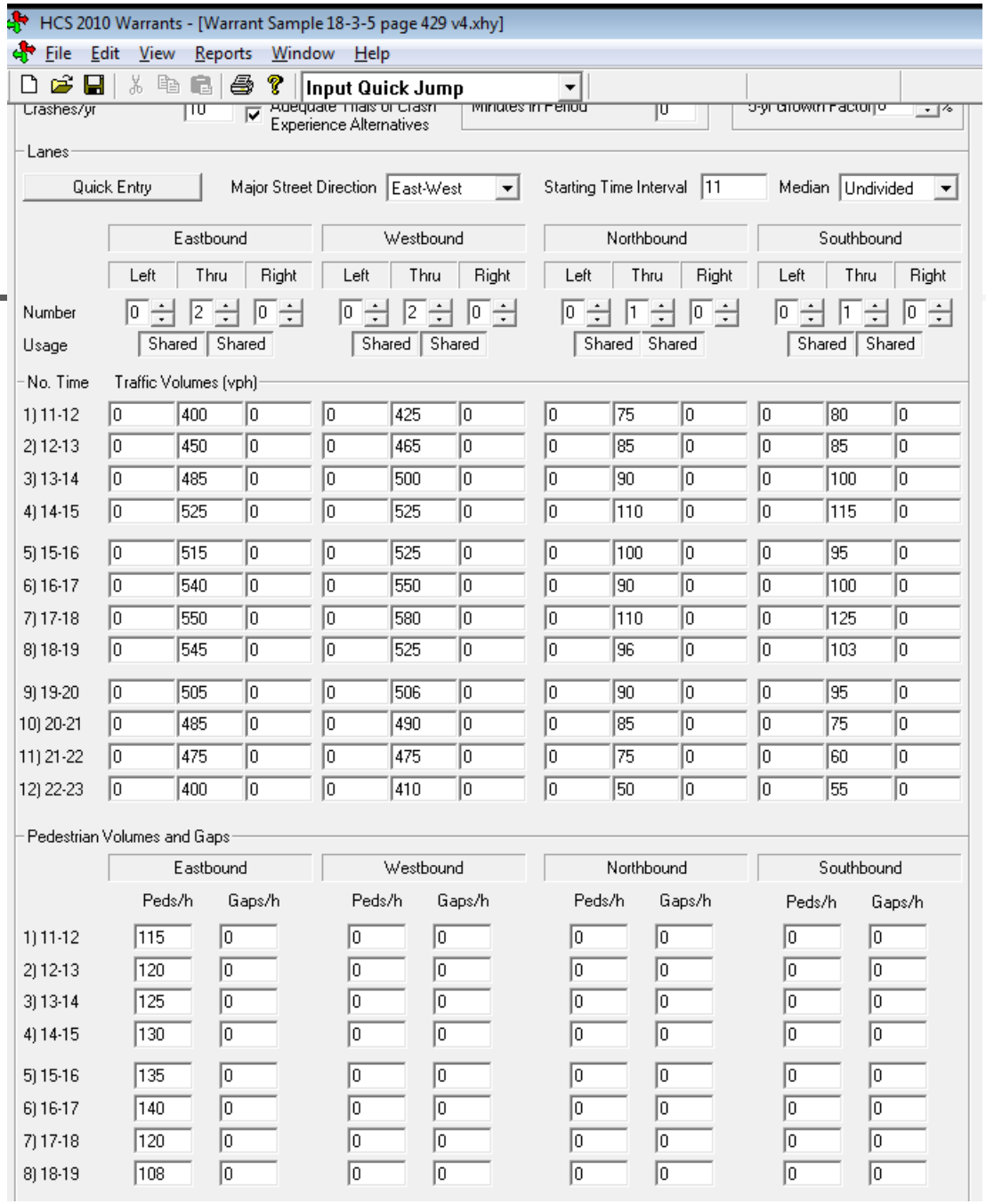
Warrants Summary

- Warrant 8: N/A
- Warrant 9: N/A

- In summary, Warrants 1B (interruption of continuous traffic), 4 (pedestrians), and 7 (crash experience) are met.
- Therefore, traffic signal control should be considered for this intersection.

How to input vehicular and pedestrian volume in HCS 2010




HCS 2010 Warrants - [Warrant Sample 18-3-5 page 429 v4.xhy]
File Edit View Reports Window Help
Input Quick Jump
Crashes/yr
Adequate Trials of Crash Experience Alternatives
Minutes in Period
3-yr Growth Factor
Lanes
Quick Entry
Major Street Direction
East-West
Starting Time Interval
11
Median
Undivided
Eastbound
Westbound
Northbound
Southbound
Left Thru Right
Number
Usage
Shared
No. Time
Traffic Volumes (vph)
1) 11-12 0 400 0 0 425 0 0 75 0 0 80 0
2) 12-13 0 450 0 0 465 0 0 85 0 0 85 0
3) 13-14 0 485 0 0 500 0 0 90 0 0 100 0
4) 14-15 0 525 0 0 525 0 0 110 0 0 115 0
5) 15-16 0 515 0 0 525 0 0 100 0 0 95 0
6) 16-17 0 540 0 0 550 0 0 90 0 0 100 0
7) 17-18 0 550 0 0 580 0 0 110 0 0 125 0
8) 18-19 0 545 0 0 525 0 0 96 0 0 103 0
9) 19-20 0 505 0 0 506 0 0 90 0 0 95 0
10) 20-21 0 485 0 0 490 0 0 85 0 0 75 0
11) 21-22 0 475 0 0 475 0 0 75 0 0 60 0
12) 22-23 0 400 0 0 410 0 0 50 0 0 55 0
Pedestrian Volumes and Gaps
Peds/h Gaps/h
1) 11-12 115 0 0 0 0 0 0 0
2) 12-13 120 0 0 0 0 0 0 0
3) 13-14 125 0 0 0 0 0 0 0
4) 14-15 130 0 0 0 0 0 0 0
5) 15-16 135 0 0 0 0 0 0 0
6) 16-17 140 0 0 0 0 0 0 0
7) 17-18 120 0 0 0 0 0 0 0
8) 18-19 108 0 0 0 0 0 0 0